

THE COMPOSITION AND DERIVATION OF APOLLO 12 SOILS

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A feldspathic orthopyroxene rich rock, with abundant Ca phosphates and K-feldspar, and chemically comparable glass fragments are found in the Apollo 12 soil. This material is rich in K, Rb, P and rare-earth elements. Three Apollo 12 soils and at least one breccia are quantitatively matched by mixtures of this material (designated KREEP) and average Apollo 12 basalt. The dark material in 12013 is similar to KREEP, and Luny Rock I (Apollo 11) is probably identical. KREEP, or similar materials, can also be the "magic" or "cryptic" component of Apollo 11 soil.

1. Introduction

The lunar samples returned from Tranquillity Base (Apollo 11) and the Surveyor III site in the Ocean of Storms (Apollo 12) consist of two types of materials: (1) igneous rocks that have crystallized from a magma qualitatively similar to terrestrial basaltic liquids, and (2) a complex mixture of glass and crystalline fragments designated as soil (or fines) and breccia (compacted soil). It has been widely assumed that the igneous rocks at both sites are samples of the bedrock on which the soil or regolith was developed. A natural, but subtle, extension of this very plausible assumption contends that the soil is predominantly derived from the underlying bedrock. However, detailed study of the Apollo 11 soil has clearly demonstrated that the soil at this site contains some components that are unrelated to the bedrock. Anorthositic fragments described by Wood et al. [1] and a single feldspathic ortho-

pyroxene rich rock (Luny Rock I) described by the Lunatic Asylum [2] and Albee and Chodos [3] cannot have come from the basaltic flows that presumably underlie the regolith. Furthermore, both the bulk chemistry and the trace-element chemistry of the soil and igneous rock samples from this site indicate that the soil must contain rock debris with a very different composition from that observed among the larger igneous rock samples [4–6]. In addition, the isotopic composition of strontium [7] demonstrates that the Apollo 11 soil contains some material that probably is substantially older than the igneous rocks at this site. Papanastassiou and Wasserburg [8] have also observed such a component – which they have designated the magic component – in the Apollo 12 soil. Thus, the lunar soil and breccia samples must carry a great deal of information about the chemistry and history of the lunar surface, information that may never be observed in the larger basaltic rock samples apparently derived

from the local bedrock. Terrestrial analogies for this situation (e.g., a glacial till) are not hard to find; however, the reconstruction of the history and chemistry of a portion of the earth's surface from such a complex sample has never been attempted. Obviously there is little motivation for such a study when more direct observations are readily accessible.

In this paper we present results on the chemical composition of common igneous rocks, several individual soil and breccia samples, and fragments selected from these samples, from the Apollo 12 site. These results show quite definitely that the presumed bedrock makes up less than half of some of the soil and breccia samples collected at this site. Thus, the foreign component for Apollo 12 is even greater than that inferred for the Apollo 11 soil.

2. Sample description and results

The results presented here consist of: (1) a series of trace-element analyses (REE, Ba, Sr, K, and Rb) and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios on four individual soil samples and milligram-sized fragments taken from breccia and soil samples, and (2) mineralogical and major-element studies on a series of milligram- and submilligram-sized glass fragments taken from Apollo 12 samples 12003 and 12033. Three different bulk soil samples from the Apollo 12 site were used in this study. They are samples 12070 and 12044 from the north and south sides of the Surveyor Crater and sample 12033 from Head Crater. Sample 12033 was taken from the bottom of a 6 in deep trench near the north rim of Head Crater [9]. It was noted at the time this sample was collected that it was lighter in color than other soil samples. This observation was subsequently verified by the preliminary examination [9]. The ejecta blanket from Head Crater apparently covered this light layer and protected it from gardening by later cratering events.

Two breccia samples, 12073 and 12013, were also analyzed. 12073 was collected with soil 12070 from the region between Surveyor and Head Crater. The location of sample 12013 is not known with certainty. It is quite likely, however, that it came from the region north of Head Crater [10].

3. General description of coarse fines

The fraction of the lunar soil that does not pass through a 1 mm sieve has been designated as coarse fines [11]. In all of the Apollo 12 soil samples, the rock types that make up this particle size range occur in very different proportions from those found for the large rocks (> 40 mm) returned from the Apollo 12 regolith. A preliminary survey of about 200 thin-sectioned particles of coarse fines (1–2 mm) indicates that they consist of approximately 60% breccia, 30% crystalline igneous rock, and 10% glass. About 4% of the crystalline igneous rock fragments in the coarse fines are either gabbroic anorthosite or a rather unusual equigranular rock made up of a mineral assemblage of orthopyroxene and plagioclase feldspar with minor amounts of potassium feldspar and rare-earth-rich calcium phosphate. This rock is similar in texture and mineralogy to a single Apollo 11 fragment, designated Luny Rock I, which was described by the Lunar Asylum [2] and Albee and Chodos [3]. We have observed five such fragments out of 200 thin-sectioned fragments in the Apollo 12 coarse fines. Five additional fragments were observed as inclusions in glass fragments, and a large number of small fragments of this material exists in the breccias. Fuchs [12] has reported observations on three additional fragments with similar mineralogy from Apollo 12. The plagioclase in the fragments described by Fuchs and in those we have observed is distinctly richer in albite (^{70}An – ^{80}An) than that reported in the anorthosites ($> ^{95}\text{An}$) from the Apollo 11 soil [1].

4. Glass fragments from coarse fines

The Apollo 12 coarse fines contain a variety of glass particles, glass spheres, and vesicular fragments similar to those found in the Apollo 11 soil. From the textures, colors, and chemical compositions determined on the Apollo 11 fragments, one can conclude that many of these fragments could have been derived from the basalts found at this site. On visual examination of Apollo 12 glass fragments, one particular type of glass fragment, a transparent yellow-brown glass, stands out as a distinctive type [13]. These fragments commonly occur as segments of ropy strands with a characteristic fluted or wrinkled surface. In other cases, they resemble

Table 1
Major-element data.

Column no.	1	2	3	4	5	6	7
Compound							
SiO ₂ (%)	48.2 ± 0.8	45.2 ± 1.5	48.2	51.7	49.0	34.0	45.0
TiO ₂ (%)	1.9 ± 0.3	3.3 ± 0.7	2.0	0.7	—	0.2	1.7
Al ₂ O ₃ (%)	15.5 ± 2.8	9.7 ± 1.6	18.0	0.8	31.1	—	21.0
FeO (%)	10.7 ± 1.8	20.1 ± 1.2	10.6	20.8	—	34.3	7.5
MgO (%)	7.8 ± 1.1	10.3 ± 3.6	7.5	22.6	—	28.8	9.5
CaO (%)	10.9 ± 0.8	10.2 ± 1.4	11.3	1.7	15.1	0.1	13.0
Na ₂ O (%)	0.45 ± 0.20	0.31 ± 0.14	0.8	—	2.8	—	0.5
K ₂ O (%)	1.2 ± 0.3	0.064 ± 0.013	0.8	—	0.05	—	0.3
P ₂ O ₅ (%)	0.7 ± 0.1	0.1 ± 0.02	0.7	—	—	—	—

1. KREEP glass, average of 22 fragments by microprobe analysis.
2. Average Apollo 12 basalt, average of 18 preliminary analyses.
3. KREEP-1, glass analysis by broad-beam microprobe.
4. Orthopyroxene in KREEP-1, average of 6 microprobe analyses.
5. Plagioclase in KREEP-1, average of 6 microprobe analyses An 73.
6. Olivine in KREEP-1, single microprobe analysis.
7. Matrix of fragment 12033, MH; broad-beam microprobe analysis.

impactite glasses with well developed flow textures and with common inclusions of feldspathic rock fragments. The feldspars in these crystalline fragments show abundant evidence of shock metamorphism (deformed twin lamellae and maskelynite). Micron-sized spheres of Fe and FeS and microlites of plagioclase are common features of the host glasses.

Electron-microprobe analyses of 22 individual fragments of this distinctive yellow-brown glass show that they are uniformly homogeneous and constant in composition. The observed scatter in the measured concentration is within that inferred from experimental uncertainties for all elements except Al and K (table 1). It is seen that these fragments are clearly and easily distinguished from the Apollo 12 igneous rocks by their chemical compositions. This distinctive glass has much higher K and P contents and also has distinctly higher Al and Ca contents and lower Fe contents. In the subsequent discussion, we will refer to this distinctive glass as KREEP glass (from its distinctive potassium, rare-earth element, and phosphorus content).

Two fragments from the coarse fines studied here have been chosen for more detailed study. The first, designated KREEP-1, was a 6.4 milligram glass fragment from soil sample 12003. The second, designated 12033, 73, MH, is a dense breccia fragment from soil sample 12033. KREEP-1 was a light brown, slightly vesicular

glass very similar to, and a member of, the KREEP glasses described above. Each fragment was split into two parts, one for thin-section and microprobe analyses (about one-third of the original fragment) and one for destructive chemical analysis. The thin-sectioned portion of KREEP-1 contained both devitrified glass and a clast of feldspathic-orthopyroxene rock. Microprobe analysis of the partially devitrified glass and the minerals in the single clast of KREEP-1 are given in table 1. In addition, accessory Fe-Ni (15% Ni), ilmenite, and rare-earth-rich calcium phosphates were observed. The microprobe analyses of this fragment show that this composite fragment is essentially identical in composition to the KREEP glasses that we have described above. Both the mineral assemblage of the clast and the composition of the minerals in the clast are very similar to those described for Luny Rock I [2] and by Fuchs [12] for three similar orthopyroxene-rich fragments from Apollo 12 soil. The occurrence of fragments with such a distinctive mineral assemblage and composition in two widely separated lunar sites is strong evidence that rocks of this composition are a significant rock type on the lunar surface.

The second fragment, 12033, MH, was also analyzed by both microprobe and destructive chemical analysis. This fragment consists of a dense breccia with inclusions of calcic plagioclase and minor quantities of clino-

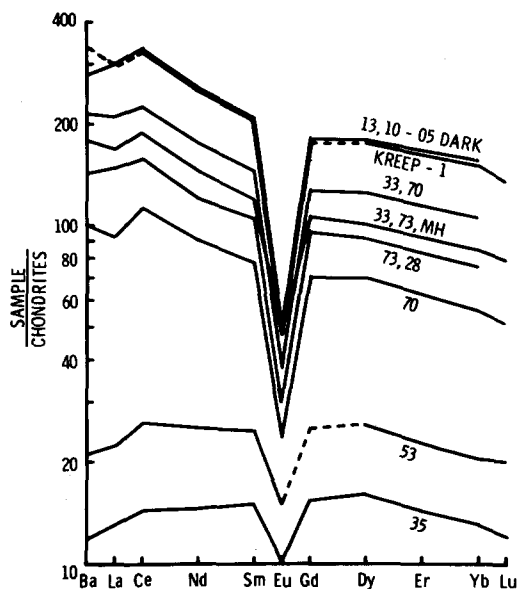


Fig. 1. Rare-earth element and Ba data for some Apollo 12 soils (12070 and 12033), breccias (12073 and 12013 dark component), fragments (KREEP-1 and 12033, 73, MH) and two typical Apollo 12 igneous rocks.

pyroxene. This fragment is similar in texture to the Apollo 12 breccia sample 12073. Breccia fragments of this type make up a major fraction of the 12033 coarse fines. Analyses of the dark opaque matrix of this fragment using a defocused electron-microprobe beam are shown in table 1. The high Al_2O_3 and CaO contents probably reflect the occurrence of An-rich plagioclase in the breccia.

5. K, Rb, Sr, Ba, and REE abundances in soil and breccia samples

Results of isotope-dilution determinations of a series of trace elements in a group of soil and breccia samples are shown in table 2 and fig. 1. These analyses were made by techniques previously described by Gast et al. [14]. Some previously reported results [15] for igneous rocks from the Apollo 12 site have very similar Ba, REE, and Sr chondrite normalized abundance patterns. However, the total abundance of these elements varies significantly. The observed REE patterns for fragment KREEP-1, (i.e. one of the high K, REE,

and P glasses) and a previously analyzed fragment of the dark portion of rock 12013 [16] are remarkably similar. The precise trace-element data presented here thus strongly support the suggestion of Hubbard et al. [16] and Schnetzler et al. [17] that the dark portion of rock 12013 may be a major component in the Apollo 12 soil. However, on observation of fragments in the soil samples, it is apparent that the KREEP glasses and their associated orthopyroxene-rich fragments are more representative of a specific rock type than the dark breccia from rock 12013. We will thus consider the hypotheses that a component of KREEP composition makes up a significant fraction of the Apollo 12 soil along with the igneous rocks of basaltic composition from this site.

In fig. 2, we have plotted the abundance patterns for an average igneous rock from the Apollo 12 site along with the abundances reported here for the KREEP glasses. The calculated abundance patterns for two mixtures of these end members are also shown. These may be compared with the patterns observed for various soil and breccia samples. In fig. 3, we have plotted the end members of a simple two-component mixing model for several trace elements and show the position of 12070, 12033, and 12073 relative to the mixing lines.

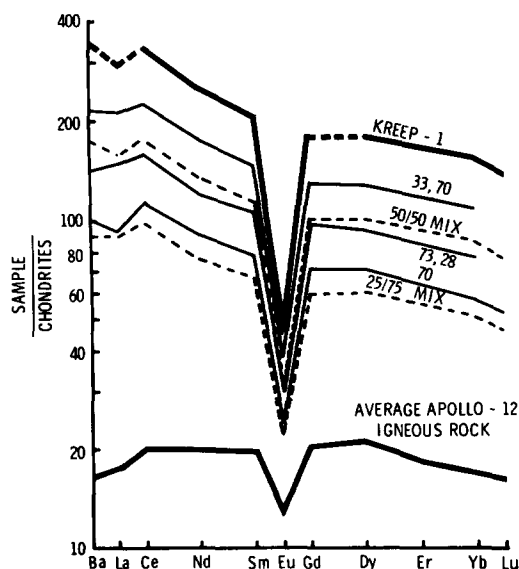


Fig. 2. Comparison of the REE and Ba values for mixtures of KREEP-1 and average Apollo 12 igneous rock with data for two soils and one breccia.

Table 2
Trace-element and isotopic data.

Sample no.	12035,16 igneous rock	12053,32 igneous rock	12044,10 soil, fines	12070,04 soil, fines	12033,70* soil, coarse fines	12073,28 breccia	12033,73, MH fragment 3.6 mg	12003 KREEP-1 fragment 4.3 mg	12013, 10-05 dark material
Element									
La (ppm)	3.87	7.32	—	30.3	69.7	48.4	55.0	—	98.7
Ce (ppm)	11.5	20.9	91.6	90.0	180	127	151	265	267
Nd (ppm)	8.91	15.3	54.8	55.1	108	76.7	89.2	155	156
Sm (ppm)	3.22	5.25	16.7	16.4	31.0	22.4	26.0	44.5	44.7
Eu (ppm)	0.751	1.10	1.72	1.73	2.79	2.17	3.46	3.67	3.38
Gd (ppm)	4.30	—	19.8	—	35.7	26.9	30.0	50.8	—
Dy (ppm)	5.07	8.01	21.2	21.7	39.2	28.6	31.5	57.0	55.8
Er (ppm)	3.09	4.88	13.3	13.5	25.1	18.9	19.7	36.0	35.1
Yb (ppm)	3.04	4.71	12.9	13.0	24.0	17.4	19.5	35.9	34.4
Lu (ppm)	0.423	0.662	1.76	1.78	—	—	2.76	—	4.66
Ba (ppm)	47.2	84.4	380	404	858	571	723	1365	1162
Sr (ppm)	84.3	130	—	150	185	163	233	216	199
$^{87}\text{Sr}/^{86}\text{Sr}^{**}$	0.7001 ±0.0003	—	—	0.7078 ±0.0003	0.7126 ±0.0003	0.7094 ±0.0003	0.707 +0.0005 -0.0015	0.716 +0.001 -0.003	0.710 ±0.0005
K (ppm)	363	582	2110	2150	4697	3180	2910	7030	3830
Rb (ppm)	0.689	1.24	6.08	6.46	14.4	9.26	8.79	20.0	9.80
K/Rb	532	469	347	333	330	347	335	356	391

* 132 mg split of 0.98 g homogenized sample of ground-up coarse fines (2-4 mm).

** Errors listed are approximately 2σ limits.

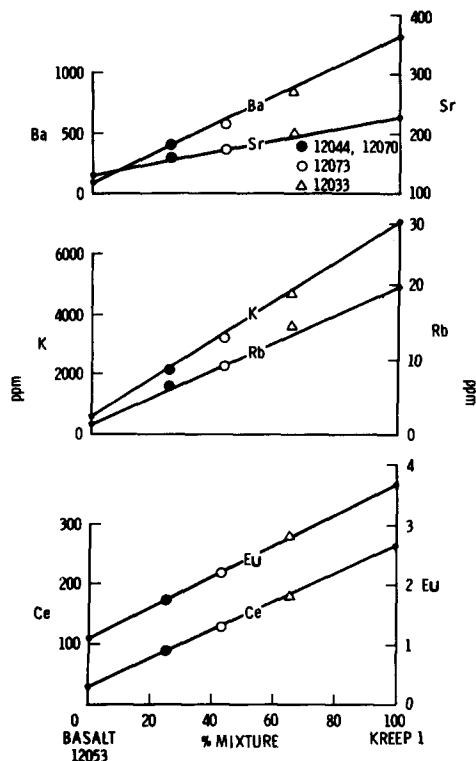


Fig. 3. Demonstration that a mixture of average Apollo 12 basalt and KREEP fits the data for several trace elements in soil samples 12070 and 12044 (25% KREEP), breccia 12073 (43% KREEP), and soil 12033, 70 (65% KREEP). Mixtures derived from REE plot.

It is seen that this rather simple two-component model can satisfy many of the soil characteristics. The required fraction of KREEP material ranges from approximately one-fourth in the Surveyor Crater Soil (12070) to nearly two-thirds in the 12033 coarse fines. The differences in proportions observed for two soil samples and the one breccia clearly indicate that the regolith at this site is not a well mixed or chemically homogenized sample.

The conclusion that a rock like KREEP makes up between one-fourth and two-thirds of the soil may appear to be in conflict with the scarcity of KREEP-like glasses in the coarse fines. This apparent discrepancy simply acknowledges the fact that the distribution of chemical characteristics in particles of the very fine dust (< 1 mm) and in the coarse fines may be quite

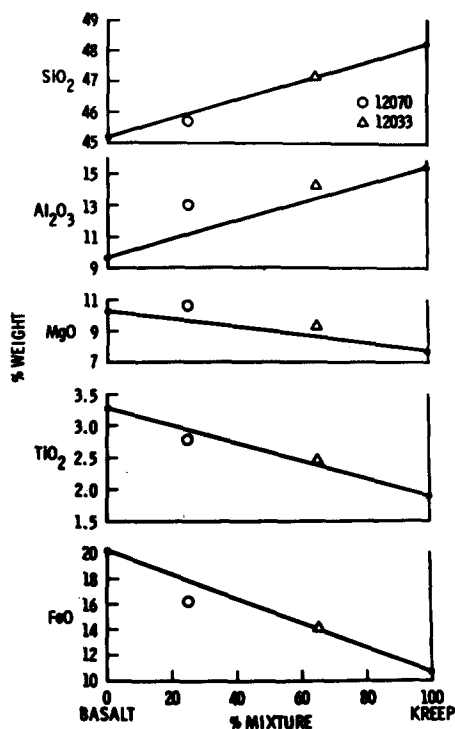


Fig. 4. Preliminary major-element data for Apollo 12 soil samples are compared with the mixing model based on trace element (fig. 3). The deviation in FeO and Al_2O_3 content of the mixture from the observed FeO and Al_2O_3 contents can be accounted for by the addition of an anorthositic component. The MgO content of the mixture cannot be precisely estimated because of the wide range in MgO concentrations of the Apollo 12 igneous rocks.

different. It is certainly not unreasonable to suppose that a component that is older and of more distant origin should be finer grained than particles derived from the underlying bedrock.

6. Major elements

The two-component mixing model for Apollo 12 soil samples, constructed using variations in trace elements, is also found to explain adequately the variations in major elements. The concentrations of six oxides in the KREEP glasses and in average igneous rocks from the Apollo 12 site are plotted as two end members of a mixing diagram in fig. 4. It should be

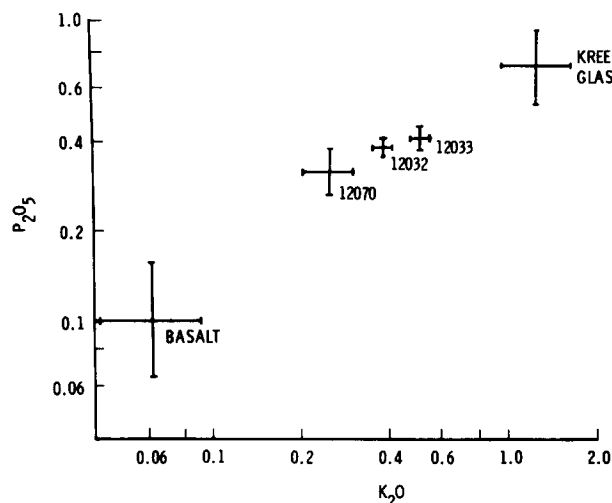


Fig. 5. The distribution of K_2O and P_2O_5 in KREEP glass, Apollo 12 soils, and average igneous rocks (preliminary data).

noted that the FeO , MgO , and Al_2O_3 concentrations in the igneous rocks cover a rather wide range (table 1). The average composition chosen here thus implies that the particular collection of igneous rock available to us is representative of the bedrock. The composition of soil samples 12070, 12044, and 12033 and breccia 12073 are plotted at the proportions inferred from the trace-element data. The agreement between the proposed mixing model and the observed concentrations in mixtures is rather good. However, the Al_2O_3 and FeO contents of the soil sample 12070 are clearly not completely accounted for. Two possible modifications can account for this discrepancy: (1) the addition of about 5% of anorthosite. This would enhance the Al_2O_3 concentration and diminish the FeO and TiO_2 content of the soil, or (2) by choosing a somewhat different composition for the basaltic end member; i.e., one of the more Al -rich and Fe -poor igneous rocks from this site. Fig. 5 shows the P_2O_5 and K_2O concentration for the basalt, soils, and breccia in relation to the values for KREEP. Significantly, in all cases the chemistry of the soil samples is a mixture of basalt and KREEP.

7. Rb-Sr systematics

Papanastassiou and Wasserburg [8] have shown that a soil sample from the Apollo 12 site gives an apparent

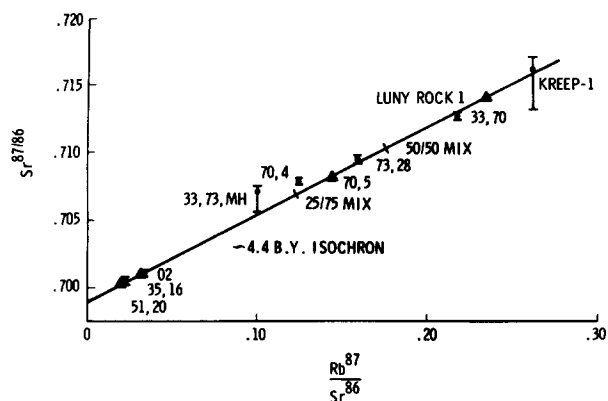


Fig. 6. An isochron plot for Apollo 12 samples and Luny Rock I. Data in solid triangles are from [7, 8]. One set of data for 12002 is unpublished data by N.J. Hubbard. The mixing points are generated from Luny Rock I and 12002 and will move up the mixing line if KREEP-1 is used, as in figs. 2.3, and 4.

Rb-Sr age of 4.4 by. In addition, the isotopic studies on rock 12013 [17, 18] show that materials probably older than 4.0 by must occur in the Apollo 12 soil. We have no direct evidence on the apparent age of the KREEP glasses studied here, although such an age is consistent with the isotopic results reported here. Using this assumption, we have plotted Rb-Sr data for the Apollo 12 samples used in this study [7, 8] and Luny Rock I (fig. 6). The data points fall on, or very near, a single line and are consistent with an apparent age of ~ 4.4 by. KREEP- or Luny Rock-like materials clearly can explain the old model age [8] for the Apollo 12 soil.

Note that the $^{87}Sr/^{86}Sr$ and $^{87}Rb/^{86}Sr$ data put the samples and mixtures in the same relative positions on the mixing line as do the rare-earth data (fig. 2) and other data (figs. 3 and 4) — except for 12033, 73, MH which also deviates in fig. 1 for Eu. Thus, the isotopic data, the trace-element data, and the major-element data can all be brought into agreement by the model proposed here.

8. Geological interpretation

The significance of the mass-balance calculations and the two-component model for the Apollo 12 soil clearly lies in the further inferences that can be drawn

from this model regarding the origin of the lunar regolith. The chemical and isotopic data cited here indicate that at the Apollo 12 site about one-third of the material that makes up the upper few inches of the soil must have originated from rocks that are much older and quite different in composition and mineralogy from the igneous rocks that apparently underlie the soil. The source of these materials is clearly of great interest and significance. It may underlie the igneous rocks which presumably cover the mare regions or may be in the highland regions. Three distinguishable hypotheses on the evolution of Apollo 12 soil can be put forth. They are: (1) the KREEP material is associated with a Copernican ray, (2) there is such widespread horizontal transport of material on the lunar surface that a substantial fraction of highland material is intermixed with the mare regolith at most points on the lunar surface, or (3) the mare flows in the Apollo 12 regions cover an earlier regolith and are so thin that they are frequently punctured by post-mare craters which mix the underlying pre-mare material with the post-mare soil.

It is clearly established that the Apollo 12 traverses are in a region that appears as a Copernican ray on the Orbiter medium-resolution photographs and on telescopic photographs [9]. Even though this feature is prominent on such photographs, it is apparently subtle enough on the lunar surface so that a human observer was unaware of the presence of such a ray. If the ray material consisted of KREEP glasses and rocks deposited to a depth of several tens of centimeters, subsequent gardening and mixing of this layer may have produced the heterogeneous mixture observed at the Apollo 12 site.

The closest source of non-mare material to the Apollo 12 site is, in fact, the Fra Mauro region. If this is postulated as the source of the KREEP glasses, we must infer that there has been very extensive horizontal transport in both directions between the Apollo 12 site and the Fra Mauro site after extrusion of the mare flows. It seems likely that such transport on the scale required would tend to obliterate some of the albedo boundaries observed on the moon; that is, sharp boundaries observed between dark and light mare material would be inconsistent with this hypothesis. The third hypothesis given above needs no further elaboration. It implies that the mare flows in the Apollo 12 region are quite thin and that there was a substantial hiatus between the development of the sub-mare regolith and the extrusion of the mare flows.

9. Apollo 11 soil

The occurrence of anorthosites and Luny Rock I in the Apollo 11 soil along with the isotopic relations cited above clearly indicates that the two igneous rock types observed at this site cannot fully account for the chemical and isotopic composition of the soil. Compston et al. [4] and Goles et al. [5] have attempted to describe the component or components that are necessary to explain the Ti, Ca, and Al content of the Apollo 11 soil. They conclude that a mixture of the high and low Rb igneous rocks with a third component can explain the observed major element compositions. Compston et al. [4] suggest that this component is chemically similar to anorthosite [1]. Goles et al. [5] do not identify a specific material for this third component. Furthermore, Tera et al. [6] have shown that the soil must contain a Cs- and Rb-rich component. It is readily shown that this component must have lower K/Rb and lower K/Cs than those observed for the high-Rb rocks if the soil is dominated by a mixture of the high and low Rb rocks. Various mixtures of returned igneous rocks, anorthosites, and an unknown component have also been postulated by several other authors. We note here that a binary mixture of $\sim 10\%$ KREEP-1 and $\sim 90\%$ of the average low-Rb rocks from the Apollo 11 site can explain the observed K/Rb correlation. Furthermore, the rare-earth abundances [14, 19–22] also show that an average of the low-Rb rocks better approximates the heavy rare-earth abundances in the soil samples than an equal mixture of high-Rb and low-Rb igneous rocks.

It is clear that any soil model based on roughly equal portions of the two igneous rock types found at the Apollo 11 site must contain at least one additional component with very unusual chemical properties; that is, low K/Rb ratios and high U contents coupled with relatively low rare-earth contents in order to explain the observed chemistry. A more consistent soil model for most of the observed chemical characteristics can be constructed from the low-Rb igneous rocks, the KREEP or Luny Rock I composition, and a small portion ($\sim 5\%$) of anorthosite. This combination of materials has the particular advantage that most of the U and Rb is associated with a component that is older than 4.0 by.

As a test of this hypothesis, we have plotted the compositions of individual glass fragments from the Apollo 11 soil that were reported in the Apollo 11 Lunar

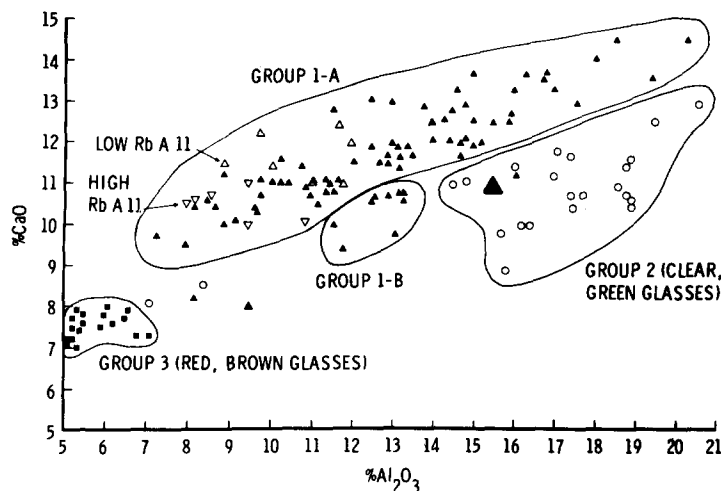


Fig. 7. CaO and Al_2O_3 content of glass fragments from the Apollo 11 soil. Data plotted are taken from all reported microprobe analyses [23]. The concentrations of other elements in Group 1-A are similar to those in the igneous rocks; Group 1-B fragments have lower TiO_2 contents; Group 2 glasses have lower TiO_2 contents ($< 6\%$) and lower FeO contents (8–12%) than the Group 1 and Group 1-A glasses. The large triangle in Group 2 is the average of the 22 analyses on KREEP glass fragments.

Science Conference Report [23] on a plot of CaO vs. Al_2O_3 contents (fig. 7). Three distinct compositional groups can be distinguished on this plot. The most common type of glass fragments have CaO and Al_2O_3 contents that overlap those of the Apollo 11 igneous rocks. However, many of the glass fragments in this group appear to be more feldspathic than the igneous rocks. The second group of glasses is the deep red and brown fragments already identified as a distinct group [24, 25]. The third group, usually described as green or clear glasses, is distinguished by distinctly higher Al/Ca ratios than those of the first group. The microprobe analyses of this third group show that these glasses are consistently low in TiO_2 , less than 5.5%, and contain between 9.0 and 12.0% of FeO. Approximately one-fourth of the Group 3 fragments have in excess of 0.3% K_2O . The 22 KREEP fragments analyzed in this study clearly overlap this third group. The identification of this distinct group of low-Ti and low-Fe glasses affords some additional direct evidence for a KREEP- or Luny Rock-like component in the Apollo 11 soil.

10. Implications

Chemical and isotopic studies of the soil at two

lunar sites strongly suggest that there exists on the surface of the moon a basaltic rock rather rich in K, P, and rare-earth elements. This rock appears to be lower in TiO_2 and FeO and higher in Al_2O_3 than the igneous rocks from both the Apollo 11 and 12 sites. Relative to the common igneous rock of Apollo 11 and 12, the chemical properties of this rock appear to be the result of extreme differentiation of an igneous liquid or a liquid produced by very small degrees of partial melting. In either case, the origin of this liquid presents an interesting problem. The chemistry of these basalts suggests that the moon may have been very extensively differentiated very early in its history. If our hypothesis regarding the origin of the KREEP component is correct, this rock appears to be a common kind of igneous rock in the upper 10 km of the moon. Even though our interpretations are based on a great deal of speculation, the hypothesis put forth here greatly enhances the significance of the gamma-ray survey of the lunar surface that has been planned for later Apollo missions. The discovery of significant contrasts in the ^{40}K activity of mare and non-mare regions on the moon seems quite likely.

Acknowledgments

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